

**Strathmore University**

@iLabAfrica Centre

DSA 8401 – Applied Machine Learning

Course Outline & Syllabus

Master in Data Science and Analytics

Semester start: Monday, 27 July 2026

Unit Information

Unit Title & Code:	DSA 8401 – Applied Machine Learning
Lecturers:	Javier Serrano & Emmanuel Bundi
Prerequisite:	Data Mining
Contact Hours:	3 hours weekly (1.5 h lecture + 1.5 h practical lab), Mondays
Office Hours:	[To be confirmed]
Email:	jgarcia@strathmore.edu - ebundi@strathmore.edu

1 Course Aim

This course provides students with a comprehensive understanding of the core concepts, algorithms, and methodologies of machine learning, enabling them to effectively analyse data, solve real-world problems, and build predictive models. Furthermore, this modernised course bridges the gap between theoretical model training and production-grade deployment, equipping students with the MLOps and Generative AI skills required to build, deploy, and monitor end-to-end machine learning systems in industry environments.

Practical work is anchored in the African technological and financial ecosystem — fintech solutions addressing under-banking, trade-corridor underwriting, and medical data modelling — so that graduates can apply these systems directly in regional industry contexts.

2 Course Learning Outcomes

By the end of this course, students will be able to:

1. **Understand** the intuition and formal foundations of modern machine learning and deep learning methods, including how, why, and when they work.
2. **Apply** machine learning techniques to solve practical, real-world problems using rigorous data engineering and feature extraction.
3. **Build, containerize, and deploy** ML models as scalable APIs using modern MLOps practices (Docker, FastAPI, CI/CD).
4. **Implement and fine-tune** modern Generative AI architectures, including Transformers, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG).
5. **Monitor** deployed models for data and concept drift, ensuring long-term reliability and fairness in production.

3 Course Content Overview

1. Fundamentals of ML & end-to-end pipelines
2. Advanced data pre-processing & feature engineering
3. Supervised learning (classical & ensemble)
4. Unsupervised learning & dimensionality reduction
5. Deep learning (FNN, CNN, RNN/Transformers)
6. Modern Generative AI (LLMs, RAG)
7. MLOps: model serving, containerization, and monitoring

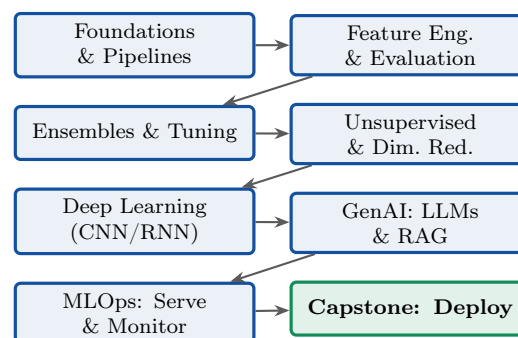


Figure 1: Course arc — from foundations to a deployed capstone system.

4 Weekly Schedule & Activities

Each session runs three hours: a 1.5-hour lecture (slide deck) followed by a 1.5-hour practical session (Jupyter/Python notebook).

Wk	Date	Topic & Intended Outcomes	Activities / Labs
1	27 Jul	Course Presentation & End-to-End ML Fundamentals — understand the ML lifecycle; set up professional dev environments.	Setup Git, VS Code, virtual envs. Build a baseline ML pipeline from scratch.
2	03 Aug	Advanced Data Prep & Feature Engineering — handle messy real-world data; prevent data leakage.	Advanced feature engineering, encoding strategies, handling missing data.
3	10 Aug	Classical ML & Rigorous Evaluation — master cross-validation and advanced metrics.	Imbalanced data (SMOTE), ROC-AUC, Precision-Recall curves.
4	17 Aug	Ensemble Methods & Hyperparameter Tuning — optimise tree-based models.	Random Forests, XGBoost/LightGBM. Automated tuning with Optuna.
5	24 Aug	Unsupervised Learning & Dimensionality Reduction — discover hidden patterns; reduce feature-space complexity.	Customer segmentation (K-Means, DBSCAN); PCA/t-SNE visualisation.
6	31 Aug	Deep Learning Fundamentals — transition from Scikit-Learn to DL frameworks.	Build and train MLPs using PyTorch/TensorFlow.

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Wk	Date	Topic & Intended Outcomes	Activities / Labs
7	07 Sep	Computer Vision with CNNs — apply transfer learning to custom image classification.	Fine-tuning pre-trained ResNet/EfficientNet architectures.
8	14 Sep	Mid-Semester Break — Capstone project proposals & dataset approvals due.	<i>No class.</i>
9	21 Sep	Sequence Models & Time Series — forecast temporal data with advanced sequence models.	LSTMs/GRUs for time series; intro to Temporal Fusion Transformers.
10	28 Sep	Modern GenAI: Transformers & LLMs — understand and apply modern Generative AI architectures.	HuggingFace ecosystem, fine-tuning LLMs, building RAG pipelines.
11	05 Oct	MLOps I: Model Packaging & Serving — wrap models in production-ready APIs; containerize.	Build a FastAPI application and containerize it with Docker.
12	12 Oct	MLOps II: Experiment Tracking & CI/CD — automate model testing; track experiments.	Integrate MLflow; build a GitHub Actions CI/CD pipeline.
13	19 Oct	Model Monitoring, Drift & System Design — detect degradation; design for scalability and ethics.	Data/concept drift detection (Evidently AI); ML system design principles.
14	26 Oct	Capstone Project: Deploy & Test — deploy end-to-end systems; conduct peer reviews.	“War room” session: cloud deployment, load testing, peer debugging.
15	02 Nov	Capstone Project Presentations — communicate technical architecture and business impact.	Live demos, architecture review, final presentations.

5 Capstone Project: Localised Themes

The capstone must deliver an **end-to-end ML system**: data pipeline → model training → API deployment (Docker/FastAPI) → frontend. Projects default to complex, messy, real-world data. Suggested tracks, anchored in the African tech and fintech ecosystem:

- **Fintech / under-banking**: alternative-data credit scoring for thin-file applicants using mobile-money (M-Pesa-style) transaction logs.
- **Trade-corridor underwriting**: risk modelling for cross-border SME lending along East-African trade corridors.
- **Health informatics**: treatment-policy modelling with Markov decision processes and reinforcement learning on clinical pathway data.
- **Fraud & AML**: streaming anomaly detection on payment rails with concept-drift monitoring.

Proposals and dataset approvals are due at the mid-semester break (Week 8).

6 Academic Assessment

Note: the traditional written final examination has been replaced with a practical midterm and a heavily weighted capstone project, to align with industry hiring standards.

Assessment Type	Weight	Description
Lab Assignments & Quizzes	20%	Bi-weekly practical labs (e.g. Dockerizing a model, building a RAG pipeline).
Midterm Exam	20%	A 2/3-hour exam focusing on data processing, model training, and evaluation.
Capstone Project — Participation & Peer Review	60%	End-to-end ML system: data pipeline, model training, API deployment (Docker/FastAPI), and a frontend. Includes active class participation and constructive peer reviews during capstone testing.
Total	100%	

7 Course Delivery Methodology

- Interactive lectures (theory & concepts) — 1.5 hours weekly, slide-deck driven.
- Hands-on practical labs (code & implementation) — 1.5 hours weekly, Jupyter/Python notebooks.
- Code reviews and peer programming.
- Comprehensive end-to-end capstone project.

8 Course Materials

Primary Textbook

- A. Géron. *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, Third Edition, 2022.

Supplementary Materials (Provided via Virtual Campus)

- C. Bishop. *Deep Learning: Foundations and Concepts*, 2024.
- C. Huyen. *Designing Machine Learning Systems* (for MLOps and system design).
- HuggingFace official documentation and courses.
- Selected research papers on Transformers, LLMs, and RAG architectures.

9 Course Policies & Guidelines

Classes

1. Punctuality is fundamental.

2. Active participation in class discussions and lab sessions is encouraged.

Assignments and Course Work

1. **Plagiarism** is a serious offence. If detected in any form in course work and assignments, the following will apply:
 - 1.a. In partial or non-serious cases (such as not citing whole word-for-word quotes), half the total possible marks of the assignment are duly struck off.
 - 1.b. In serious cases (such as whole duplication of a paper), a zero policy will apply, i.e. all offending assignments will be awarded a mark of zero.

Note: the level of seriousness referred to above is at the discretion of the lecturer. Appeals are possible through the relevant channels.

2. **Referencing:** APA system.
3. Notwithstanding the above, **collaboration** in course work is certainly encouraged as this promotes team spirit and group synergy, provided originality is preserved.

10 Communication Channels

1. **Virtual Campus:** all course materials, announcements, and assignment submissions are handled here.
2. **Institutional email:** for official communication and queries.

First session: Monday, 27 July 2026.

Please bring a laptop with Git installed and a working Python environment.